POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks:100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.	a)	AI is taking over the jobs of many Professionals in IT sector. How do you analyze the impact of AI in the industry and education?	7
	b)	Define intelligent agents. Explain the goal based agent with suitable example.	8
2.	a)	What is a Constraint Satisfaction Problem (CSP)? Solve the following cryptarithmetic problem: LOGIC + LOGIC = PROLOG	7
	b)	Explain how A-star search finds the optimal solution when there is more than one solution in the graph.	8
3.	a)	How does alpha-beta pruning improve the efficiency of minimax search? Explain with example.	7
	b)	Given the propositional statements p: "It is raining" and q: "The road is wet," write the logical expressions for the following: i. If it is not raining, then the road is not wet. ii. If it is raining, then the road is wet, and if the road is wet, then it is raining. iii. It is not the case that both it is raining and the road is wet. iv. If the road is wet, then it is raining, but if it is not raining, then the road is not wet.	8
4.	a)	What is the purpose of Baye's theorem in reasoning? Consider the following facts: i. Everybody loves someone. ii. All the dogs has a tail. iii. People try to assassinate a ruler who is not loyal to. Represent these facts in predical logic and prolog statements.	8
	b)	Define support vectors. What is the significance of maximizing the margin in SVM? Explain in detail.	7

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OR

Define reinforcement learning and explain the following components of an RL problem: state, action, reward, and policy.

5. a) Write down the algorithm for K-means clustering. Cluster the given eight data points into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9). 8

   b) What is an Artificial Neural Network? Explain how a neuron is fired or not assuming your own input values, weights and a activation function. Draw the neuron diagram and show the calculation details. 7

6. a) Explain how the single layer perception learns with suitable example. 8

OR

What is Hopfield Network? Explain in detail.

   b) How can ambiguity occur at phonetic, syntactic, semantic and pragmatic levels of natural language processing? Give examples. 7

7. Write short notes on: (**Any two**) 2×5

   a) Deep learning
   b) Expert System
   c) Semantic Nets

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